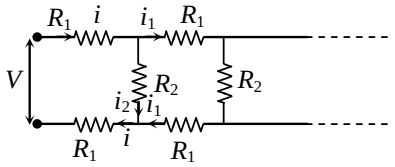
PAPER-I
PHYSICS
SECTION – I
Single Correct Choice Type

1. $i = i_1 + i_2$ …(1)

$\frac{V}{n} = 2V_{R_1} + \frac{V}{n^2}$ …(2)

$V = 2V'_{R_1} + \frac{V}{n}$ …(3)



$V_{R_1} = \frac{V}{2n^2}(n-1)$

$V'_{R_1} = \frac{V(n-1)}{2n}$

Using equation (1) $\frac{V}{2nR_1}(n-1) = \frac{V(n-1)}{2n^2R_1} + \frac{V}{nR_2}$

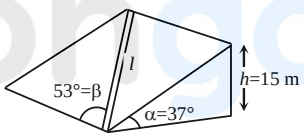
On solving $\frac{R_1}{R_2} = \frac{(n-1)^2}{2n}$

∴ **(b)**

2. $l = \frac{25}{12}h$ …(1)

$a = (g\sin 37^\circ \cos 37^\circ) - (\mu g \cos 37^\circ + \mu g \sin^2 37^\circ)$

$a = \frac{62}{25}$ m/s^2 …(2)



Use $l = \frac{1}{2}at^2$

$t = 5$ sec.

∴ **(a)**

3. For a body to float average density of body should be less than or equal to density of liquid.

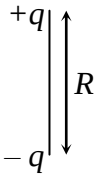
∴ **(d)**

4. $\vec{\tau} = I\vec{\alpha}$

$qRE - fR = \frac{2}{3}mR^2\alpha$ …(1)

$f = ma$ …(2)

$a = R\alpha$ …(3)



On solving above equations,

$f = \frac{3}{5}qE$

∴ **(b)**

5. After a long time the velocity of both the conductors become same.

∴ **(b)**

6. Using conservation of flux

$$L_1\frac{2E}{R} = L_1 i + L_2 i$$

$$i = \frac{L_1\dfrac{2E}{R}}{L_1 + L_2}$$

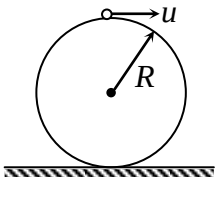
$\therefore$ **(a)**

7. Volume of cube = $(8)^3 = 512\ \text{cm}^3$

As the result should have one significant digit so it should be $500\ \text{cm}^3$.

$\therefore$ **(b)**

8. $mg - N = \dfrac{mu^2}{R}$



If $N \to 0 \Rightarrow u \to \sqrt{gR}$

So range is $0 \le u < \sqrt{gR}$

$\therefore$ **(d)**

SECTION – II
Multiple Correct Choice Type

9. For isochoric process $\dfrac{P}{T}$ = Constant

For adiabatic process $P^{1-\gamma}T^{\gamma}$ = Constant

$$\frac{dP}{dT} = \frac{\gamma}{(\gamma-1)}\left(\frac{P}{T}\right)$$

$\therefore$ **(b, c)**

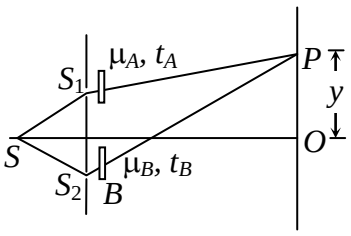
10. Path difference = 0 for central maxima

$$\{SS_2 + (S_2P - t_B) + \mu_B t_B\} - \{SS_1 + (S_1P - t_A) + \mu_A t_A\} = 0$$

$$(S_2P - S_1P) = -\mu_B t_B + t_B - t_A + \mu_A t_A$$

$$y\frac{d}{D} = (\mu_A t_A - \mu_B t_B) + (t_B - t_A)$$

Shift, $y = \dfrac{D}{d}\left[(\mu_A t_A - \mu_B t_B) + (t_B - t_A)\right]$



(i) if $t_B = t_A \Rightarrow y = 0$

(ii) if $t_B > t_A \Rightarrow y > 0$

(iii) if $t_B < t_A \Rightarrow y < 0$

$\therefore$ **(a, b, c)**

11. $E_{K\alpha} = \frac{3}{4}RhcZ^2 = \frac{3}{4}x$, where $x = RhcZ^2$

$E_{L\beta} = \frac{3}{16}x$

$E_{K\beta} = \frac{8}{9}x$

$E_{m\alpha} = \frac{7}{144}x$

$E_{m\gamma} = \frac{15}{16}x$

Now, $E_{k\alpha} + E_{L\beta} = \frac{15}{16}x$

$E_{k\beta} + E_{m\alpha} = \frac{15}{16}x$

$\therefore$ **(a, c, d)**

12. $\frac{1}{y} + \frac{1}{x} = \frac{1}{f}$

$-\frac{1}{y^2}V_i - \frac{1}{x^2}V_0 = 0$

x-direction $V_i = -\left(\frac{y}{x}\right)^2 V_0 = -\left(\frac{-60}{-60}\right)^2(-2\hat{i}) = 2\hat{i}$ cm/s

$m = -\frac{(-60)}{(-60)} = \frac{y'}{x'} \Rightarrow \frac{dy'}{dt} = \hat{j}$ cm/s

$\therefore$ **(a, d)**

SECTION–III
Comprehension Type
Passage – 1

13. $2mV_0 = 3mV \Rightarrow V = \frac{2V_0}{3}$

Impulse = $2m\left(\frac{2V_0}{3}\right) - 2mV_0 = -\frac{2}{3}mV_0$

$\therefore$ **(b)**

14. $KE_{\min} = \frac{1}{2} \times 2m\left(\frac{2}{3}V_0\right)^2 + \frac{1}{2}m\left(\frac{2}{3}V_0\right)^2 = \frac{2}{3}mV_0^2$

$\therefore$ **(a)**

15. $-1=\frac{V_1-V_2}{V_0-0}$ …(1)

$2mV_0=2mV_1+mV_2$ …(2)

$V_1=\frac{V_0}{3};\ V_2=\frac{4}{3}V_0$

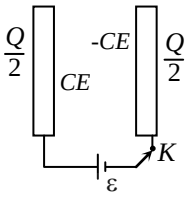
$$\frac{K_1}{K_2}=\frac{\frac{1}{2}2m\left(\frac{V_0}{3}\right)^2}{\frac{1}{2}m\left(\frac{4}{3}V_0\right)^2}=\frac{1}{8}$$

$\therefore$ **(d)**

Passage – 2

16. Charge flown $=\left(CE+\frac{Q}{2}\right)-Q$

$=\left(CE-\frac{Q}{2}\right)$



$\therefore$ **(d)**

17. Charge on the inner surface of left plate should be CE

$\therefore$ **(d)**

18. Work done = $\Delta qE=\left(\frac{3Q}{2}\right)\left(\frac{2Q}{C}\right)=\frac{3Q^2}{C}$

$\therefore$ **(b)**

SECTION–IV
Matrix–Match Type

1. **(A) – (1), (2), (4); (B) – (1), (3), (5); (C) – (2), (4); (D) – (5).**

2. **(A) – (2), (5); (B) – (1), (5); (C) – (3), (4); (D) – (3), (4).**